Framework
Focus Session

Guided Discussion

DOMMx – Data Operating Model & Maturity

An Integrated Operating Model for Assessment, Governance Execution, and Continuous Capacity Improvement

LEONARDO GUERREIRO, JOSÉ MARTINS, MARIA DO ROSÁRIO BERNARDO, HENRIQUE MAMEDE, AND FREDERICO BRANCO

Executive Summary

Organizations increasingly rely on data as a strategic asset, but they continue to face difficulties in transforming Data Governance initiatives into effective, sustainable, and operationalized capabilities. Despite the increase in investments in data, analytics, and technological platforms, there is a recurring gap between maturity diagnosis, decision-making, and practical execution, aggravated by regulatory pressures, audit requirements, accountability responsibilities, and, more recently, by the implications of the adoption of Artificial Intelligence.

The literature and organizational practice show that many Data Governance initiatives fail not because of the absence of reference frameworks or technological solutions, but because of the lack of an operating model that connects, in a structured and traceable way, maturity assessment, governance mechanisms, and concrete actions for organizational evolution. Overly descriptive models, approaches focused only on regulatory compliance, or initiatives that are heavily dependent on tools tend to result in programs that are fragmented, low adoption, and difficult to sustain over time.

In this context, this work proposes the DOMMx – Data Operating Model & Maturity, an operating model designed to integrate and operationalize existing governance frameworks, linking maturity, decision and execution. DOMMx does not replace consolidated frameworks, but acts as a superior operational layer that orchestrates them into a coherent system of evaluation, decision, and action, oriented to the incremental evolution of organizational capabilities.

The model was developed in the context of scientific research, based on a systematic review of the literature, and aligned with widely recognized references, such as DAMA-DMBOK, DCAM and organizational maturity principles inspired by CMMI. DOMMx structures governance based on a multidimensional assessment by domains, based on operational evidence, associated with a decision logic that translates maturity levels into standardized actions, procedures, prerequisites, and auditable deliverables.

By integrating governance, compliance, auditing, and organizational evolution, DOMMx recognizes that Data Governance is no longer an optional good practice and has become an indispensable organizational capability. This framework becomes particularly relevant in view of the emerging requirements of Artificial Intelligence Governance, in which traceability, explainability, risk management and responsible use of data depend on solid data governance foundations.

This executive summary presents the general framework of DOMMx and introduces the model that will be discussed and analysed in the context of an Expert Focus Session, with the participation of experts from industry and academia, with the aim of validating the coherence, practical applicability and opportunities for evolution of the model.

Table of Contents

1 Context and problem 3
2 What is DOMMx? 3
3 Contextualization 4
4 How does DOMMx work? 4
4.1 DOMMx Maturities Scales 5
4.2 Actual structure of the DOMMx die 5
4.3 Complete example with real procedures: 6
5 Where does DOMMx best apply? 7
6 What is DOMMx not? 7
7 Objetivo da Expert Focus Session 7
8 Roadmap of DOMMx 8
9 Final considerations 8
10 Authorship and Contributions 10

1 Context and problem

Despite significant increases in investments in data, analytics, and technology platforms, many organizations continue to face difficulties in making data governance effective, sustainable, and operational. In practice, Data Governance initiatives often fail not because of a lack of reference tools or frameworks, but because of the absence of an operating model that connects, in a clear and structured way, maturity diagnosis, decision-making, and practical execution.

In parallel, the intensification of data protection and privacy laws, increased audit requirements, and growing pressure for regulatory compliance have structurally altered the role of governance in organizations. What was previously treated as a good practice or a competitive differential is no longer a should have and has become inevitable and indispensable. The absence of clear governance mechanisms exposes organizations to legal, financial, reputational, and operational risks, often identified only at critical moments, such as audits, security incidents, or regulatory sanctions.

In this context, there is a recurrent fragmentation between:

- Maturity models that are predominantly descriptive, with low action orientation;
- Approaches that are overly focused on regulatory compliance, handled reactively;
- Initiatives heavily dependent on specific tools;
- Governance structures that do not evolve consistently with the organizational context.

This scenario results in governance programs that are difficult to sustain over time, with a low level of adoption, high organizational effort, and limited impact on the business. While data governance is increasingly recognized as a critical organizational capability, many initiatives continue to lack an operating model that integrates maturity assessment, execution mechanisms, and continuous capability improvement in a coherent and pragmatic way.

2 What is DOMMx?

DOMMx – Data Operating Model & Maturity is a data governance operating model, designed to integrate and operationalize existing reference frameworks, explicitly linking maturity assessment, decision-making, and practical execution. Its objective is to support organizations in building, executing and continuously evolving real data governance capabilities, in a structured, progressive and sustainable way.

The suffix "x" represents the operational, cross-sectional, and executable dimension of the model, indicating that DOMMx is not limited to the description of maturity levels, but explicitly connects diagnosis, decision, and action, allowing different governance domains to evolve in a coordinated manner over time.

DOMMx was developed in the context of scientific research, based on a systematic review of the literature, and aligned with widely recognized reference frameworks, such as DAMA-DMBOK, DCAM, and organizational and process maturity models, incorporating CMMI principles, particularly with regard to incremental capacity evolution, institutionalization of practices, and continuous improvement. At the same time, the model was designed with an explicit focus on practical applicability, tool independence, and adaptation to different organizational contexts.

In addition, DOMMx recognizes that data governance is no longer just a good practice and is now heavily constrained by data protection and privacy laws, audit requirements, accountability responsibilities, and regulatory compliance pressures. This framework becomes even more critical in the face of the emerging implications of Artificial Intelligence Governance, where automated decisions, traceability, explainability, and risk management require solid data governance foundations, supported by consistent operating models.

3 Contextualization

Data Governance refers to the set of structures, roles, processes, and mechanisms that guide how data is managed, used, and controlled throughout its lifecycle. Unlike technical data management, governance establishes responsibilities, decision criteria, and control mechanisms, ensuring alignment with organizational objectives, regulatory requirements, and accountability expectations.

To make governance operational and structurable, reference frameworks have consolidated the concept of governance domains, which represent distinct but interdependent areas of capability. Organisation by domains makes it possible to deal with governance in a systematic way, avoiding isolated initiatives and facilitating coordination between responsibilities, processes and controls.

In the context of DOMMx, data governance is analyzed from an integrated set of domains, which include, but are not limited to:

- **Data Governance**, as an integrating domain of principles, policies, roles and decision-making structures
- **Data Quality**, focused on defining, monitoring and improving data quality
- **Metadata Management**, responsible for the meaning, classification and traceability of data
- **Data Architecture** and **Data Integration & Interoperability**, related to data structure, flows, and interoperability
- **Data Security and Data Privacy & Compliance**, responsible for protection, confidentiality, proper use and legal compliance
- **Master & Reference Data**, focused on the consistency of critical and shared data
- **Analytics & Business Intelligence**, which consume governed data for decision support
- **Data Lifecycle & Retention**, associated with data retention, archiving, and disposition
- **Data Ethics & Responsible Use** and **AI & Advanced Analytics Governance**, which address the responsible use of data and the implications of adopting AI-based solutions

Although described separately, these domains do not operate independently. The effectiveness of governance depends on coordination between domains, clarity of responsibilities and the existence of mechanisms that allow its joint evolution over time.

Data governance is also understood as a capability that evolves in maturity levels, reflecting the degree of formalization, institutionalization, and effectiveness of the practices adopted. This maturity logic allows assessing the current state of governance capabilities and guiding evolution decisions in an incremental and contextualized way.

This contextualization establishes the conceptual basis necessary for understanding the operational model presented in the following sections, without prescribing specific solutions or anticipating execution mechanisms.

4 How does DOMMx work?

DOMMx works as a matrix operating model, in which each element evaluated follows a direct and traceable logic. There is no subjective interpretation or generic recommendations. The model defines what to evaluate, how to score, and what to do next, consistently across all domains and all integrated with the main concepts and practices existing in the literature and in the market, including connecting directly with the frameworks and their respective chapters.

Question → Score → Action Code → Procedures → Evidence

In DOMMx:

- The question defines the governance mechanism evaluated
- The score defines the specific Action Code
- Each Action Code activates a standard set of procedures
- There are no generic action classes (FA, SA, etc.) in the final artifact
- The mapping is Score → Action Code of the domain, according to the documented decision trees

Template literal example:
Score 3 → DG-04 (Embed governance into organizational and domain routines)

4.1 DOMMx Maturities Scales

Score	Significado
0	Capacidade inexistente
1	Prática inicial, informal
2	Parcial, não institucionalizada
3	Definida e repetível
4	Integrada, monitorizada
5	Otimizada e em melhoria contínua

4.2 Actual structure of the DOMMx die

- **Domain**: Data Governance **(DG)**
  - **Question 1 - (DG-Q1):**
    - ✓ ***Is there a corporate Data Governance policy?***

Score	Action Code (DOMMx)	Objetivo do Action Code	Procedimentos acionados
0	DG-01	Definir princípios e scope básicos de governança	Definir princípios, scope, rascunho de política, papéis iniciais
1	DG-02	Atribuir ownership e responsabilidades	Formalizar roles, stewardship e autoridade
2	DG-03	Aprovar regras e estruturas de governança	Criar council, aprovar policy e standards
3	DG-04	Integrar governança em rotinas organizacionais	Embutir checkpoints, stewardship routines
4	DG-06	Automatizar validação e monitorização	Controles automáticos, dashboards
5	DG-09	Melhorar continuamente a governança	KPIs, auditorias, ciclos de melhoria

Mapping described in DOMMx tables 27–29

4.3 Complete example with real procedures:

Action Code: DG-04 – Integrate governance into organizational and domain routines.

1. **Procedure: Integrate governance into organizational and domain routines**

   ✓ **Prerequisite:**

   It requires approved governance principles, defined scope, and formal assigned ownership and stewardship roles.

   ✓ **Deliverable:**

   Governance routines incorporated into operational processes.

   ✓ **Recommendations:**

   - Integrate governance checkpoints into business processes and data lifecycles, in alignment with DAMA's guidance on the operationalization of decision frameworks in Data Governance (DMBOK2, Chapter 3.4, p. 60–63).
   - Integrate stewardship activities into day-to-day operational routines, following DAMA's recommendations for stewardship execution and accountability (DMBOK2, Chapter 4.3, p. 90–94).
   - Align governance routines with DAMA's guidelines for integrating policies and standards into operational processes, ensuring that governance decisions are applied through existing workflows (DMBOK2, Chapter 3.5, p. 64–66).
   - Define formal control, validation and review points, consistent with the governance control and escalation mechanisms described by DAMA (DMBOK2, Chapter 3.6, p. 67–69).
   - Ensure that governance routines are documented and executed consistently across domains, in line with DAMA's guidance for institutionalized governance practices (DMBOK2, Chapter 3.7, p. 70–72).

Note: Use DCAM practices related to the integration of governance into *business-as-usual* processes, ensuring that governance execution is aligned with existing operating models and control environments.

Note: Apply CMMI institutionalization principles to confirm that governance routines are not ad hoc, but repeatable, monitored, and incorporated into standard organizational practices.

5 Where does DOMMx best apply?

DOMMx is best applied in contexts where the organization needs to make structured decisions about data governance, based on real maturity, and turn those decisions into concrete execution, in a progressive, auditable way.

In particular, the model shows greater value in the following scenarios:

- Initiating or restructuring Data Governance programs, in centralized, federated or hybrid models, where it is necessary to define priorities and avoid generic or overly ambitious approaches.

- Maturity assessments with a practical purpose, such as support for audits, regulatory compliance requirements or continuous improvement plans, when the objective is not only to classify, but to guide concrete actions.

- Growing or transforming organizations, where data complexity increases faster than governance capacity, requiring clear decisions about where to structure, formalize, or institutionalize practices.

- Contexts of organizational change, such as mergers, acquisitions or reorganizations of data and technology, where it is necessary to harmonize practices, clarify responsibilities and establish a common basis of governance.

DOMMx can be applied to organizations of different sizes and sectors, as long as there is an explicit intention to treat data governance as an evolving organizational capability, and not as a one-off initiative, dependent on tools or isolated efforts.

6 What is DOMMx not?

To avoid misinterpretations, it is important to note that DOMMx:

- Not a technological tool
- Not a compliance checklist
- Not a one-size-fits-all generic or prescriptive model/framework
- It is not a substitute for strategic or contextual decisions

The model acts as a diagnostic and guidance tool, supporting informed decisions, but not automating or rigidly standardizing governance.

7 Objetivo da Expert Focus Session

The Expert Validation Session aims to promote a critical, informed and constructive analysis of DOMMx, bringing together experts from industry and academia to discuss its conceptual coherence, methodological soundness and operational feasibility.

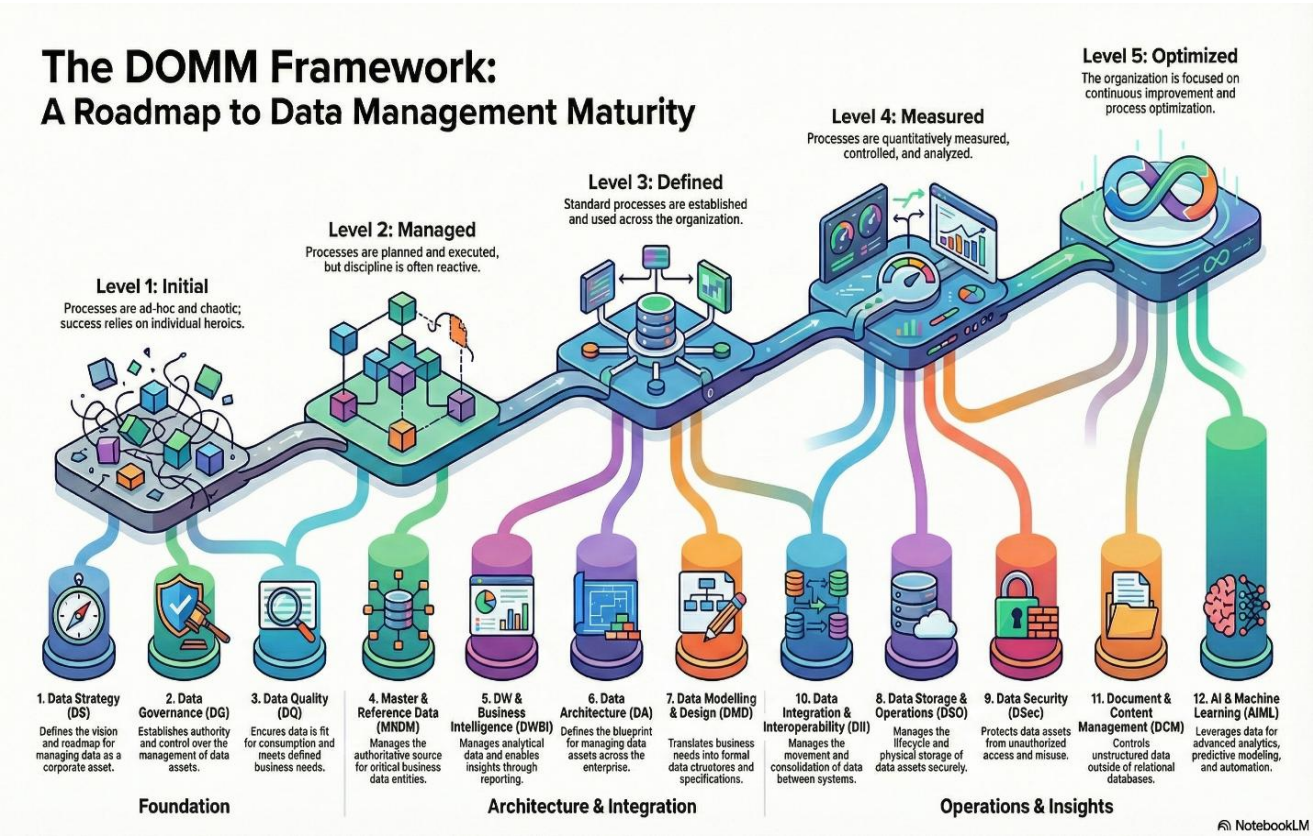
The session is designed as a structured debate, based on previously made available material, in which participants are invited to evaluate the model not only from a theoretical point of view, but also in the light of practical experiences, real organizational contexts and current data governance challenges.

In particular, the session aims to assess:

- The conceptual clarity of the model and the coherence between its components
- The theoretical adherence and methodological rigor underlying its construction
- The practical applicability of the model in different organizational contexts
- The limitations, risks and assumptions associated with its adoption
- The opportunities for improvement, extension or refinement of the model

The feedback collected during the session will be used for the final refinement of DOMMx and for the consolidation of its scientific validation process, ensuring that the model maintains academic consistency without losing practical relevance.

8 Roadmap of DOMMx



9 Final considerations

DOMMx proposes an integrated approach to data governance, seeking to reduce the gap between theory and practice. This executive summary is intended to support the initial reading of the template and to prepare participants for a critical and in-depth discussion during the focus session.

10 Authorship and Contributions

LEONARDO GUERREIRO with a solid academic background and a consolidated career in the industry, as well as experience as a teacher and instructor. The academic and professional trajectory is marked by significant achievements in the fields of information technology, engineering, and data governance. Technician in Data Processing from ISA, and graduated in Management Informatics from Universidade Nove de Julho, he started a postgraduate degree in IT from the Federal University of Paraná and obtained an MBA in IT Governance from SENAC Paraná, all in Brazil. In Portugal, he started his master's degree in software engineering at the Polytechnic Institute of Setúbal and, currently, there is a PhD search in Web Science and Technology at the University of Trás-os-Montes and Alto Douro, with a thesis under development in the area of Data Governance. It also has training courses aimed at DAMA's CDMP certification and EXIN's GDPR.
In the academic sphere, there was five years of experience as an assistant professor in graduate courses at renowned institutions in Brazil, teaching subjects related to data and information. Publication in Information Systems and Technologies. In the sector, there are more than two decades of experience working and leading complex and innovative projects internationally, recognized for tangible results and lasting impact in the field of information technology and strategic data.

FREDERICO BRANCO is an assistant professor at the University of Trás-os-Montes and Alto Douro and a senior researcher at INESC TEC. He currently serves as Pro-Rector for Information Systems. He has published more than 120 articles in journals and proceedings of events. He is also involved in various academic works as a project advisor, dissertation and thesis, and continuously participates in various research projects. His professional career is also directly related to industry, focusing on various planning and implementation projects in information systems, with special attention to the agri-food, mobility and services sectors. Throughout his career, he has held different senior management positions in operations, information systems and quality management.

MARIA DO ROSÁRIO BERNARDO holds a PhD in Management from the Higher Institute of Economics and Management of the University of Lisbon (ISEG-UL). She also holds a degree in Economics and a master's degree in mathematics applied to Economics and Management from ISEG-UL.
Since 1997, she has been a prominent professor at the Open University (UAb), teaching undergraduate, master's and doctoral courses in Economics, Management and Information Systems. She has supervised several master's and doctoral dissertations in these areas. At UAb, she held several key positions, including Coordinator of the Bachelor's degree in management (2016-2019), Master's degree in e-commerce and Internet (2014-2015) and Master's degree in management (2009-2011). She served as Vice-President of the Scientific Council (2013-2015) and as a member of the Coordinating Council of the Department of Social Sciences and Management (2013-2015 and 2019-2020). She is currently the Coordinator of the Quality Group of the Management Section of the DCSG (2012-2014 and 2020-present).
She is also a professor in the PhD Program in Web Science and Technology, a joint program between UAb and UTAD. Since 2016, she has been the Scientific Coordinator of the "Management and Economics" Community of the Open Repository of the Open University.
His contributions go beyond the university. She has been a member of the scientific committee of several national and international conferences and has served on the scientific committee of several prestigious journals.

JOSÉ MARTINS is currently a Guest Lecturer at the Polytechnic Institute of Bragança, as well as a member of the INESC TEC Research Centre.
He has published more than 100 articles in indexed journals and proceedings of events. He is also the supervisor of several master's and doctoral dissertations. During his research career, he has participated in several research projects and is currently part of research and development projects aimed at offering innovative solutions to digitally promote and develop rural and low-density regions and their endogenous products. During his professional career, he also served as a Senior Consultant for Information Systems and Technologies, participating directly in several international projects. He also dedicates most of his time to lectures and research activities, where he tries to understand the variables and (in)indirect impacts of ICT adoption at individual and company levels to drive the digital transformation of both companies and regions and foster the ICT4Development theme.